

DEF CON 26
2018-08-10



**PLEASE
DO NOT
DUPLICATE**

Attacking the Knox Box and Other Keyed Alike Systems

m010ch_

?



@m010ch_



zamak@protonmail.ch

Key boxes



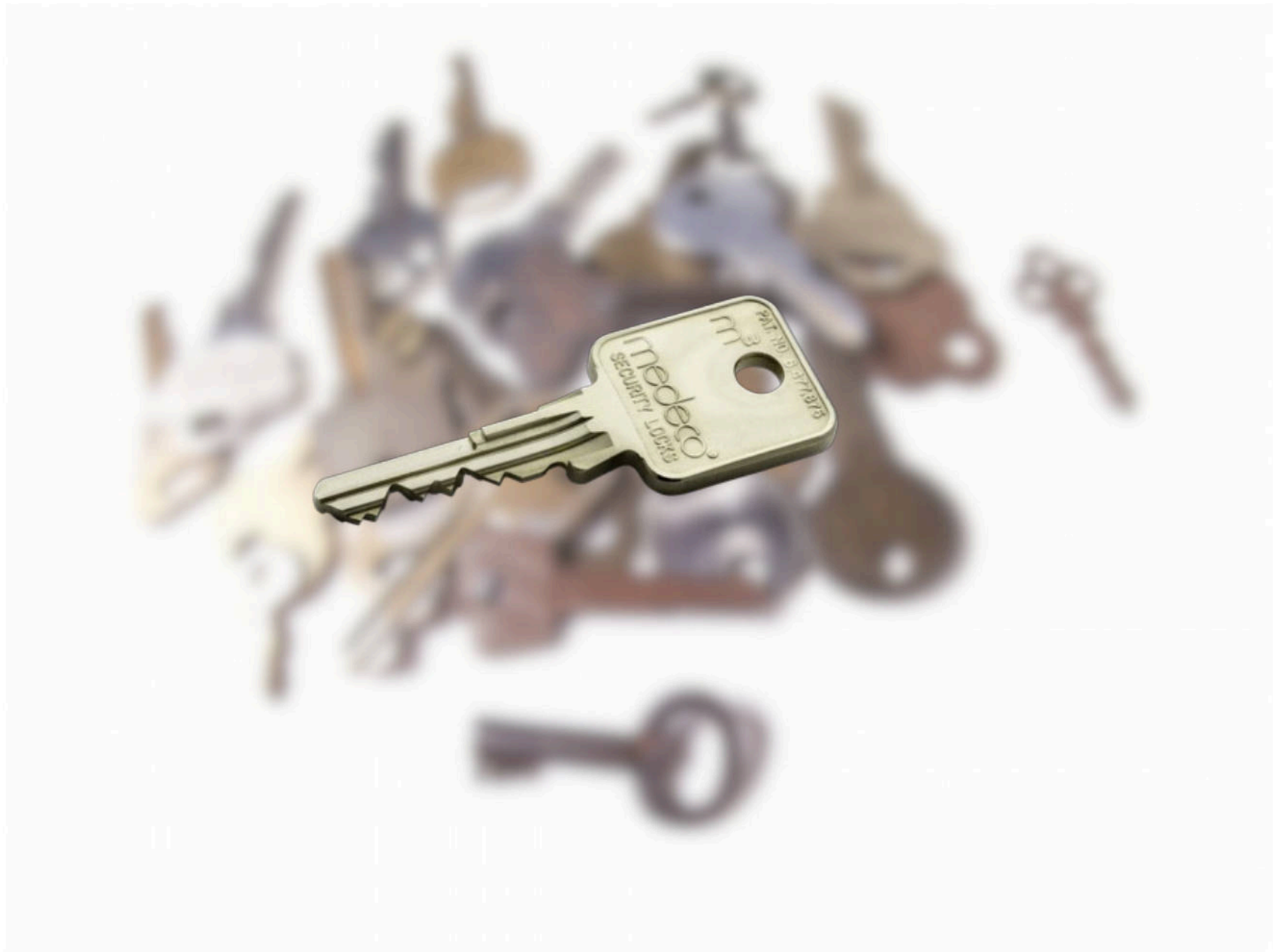
Key boxes



Key boxes



Key boxes



Who makes them?



KNOX®

When seconds matter™

Knox Boxes



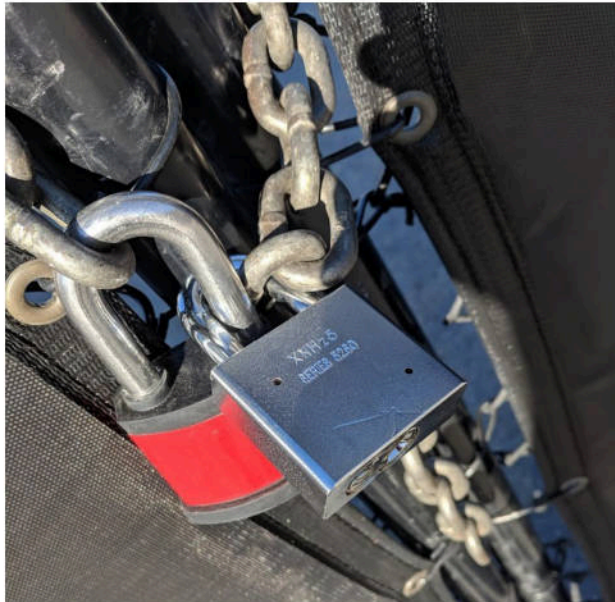
Knox Boxes



Knox Boxes



Knox Boxes



Knox Boxes



One key



05/29/2016 04:00:48 AM

CH9



05/29/2016 04:00:53 AM

CH9



05/29/2016 04:08:13 AM

CH9



05/29/2016 04:08:19 AM

CH9



05/29/2016 04:08:43 AM

CH10



06/03/2016 03:26:20 AM

CH10



06/03/2016 03:26:20 AM

CH10

Q13 FOX

[NEWS](#)

[THIS MORNING](#)

[LINKS](#)

[SPORTS](#)

[WMW](#)

[PICK MY PET](#)

[APPS](#)

[SCHEDULE](#)

[TRAFFIC](#)

[WEATHER](#) ☁ 59°

Seattle spending \$500,000 to re-key lock boxes after burglaries

POSTED 4:43 PM, OCTOBER 3, 2013, BY [STEVE KIGGINS](#), UPDATED AT 05:24PM, OCTOBER 3, 2013





NEWS ▾ WEATHER ▾ TRAFFIC ▾ INVESTIGATIONS REPORT IT SPORTS ▾ STUDIO 512

Austin

Security at Austin buildings 'compromised' after burglary suspect gets master key

By: Calily Bien 

Posted: Jan 09, 2018 10:10 AM CST

Updated: Jan 09, 2018 10:10 AM CST



[citation needed]



Only Your Fire Department Holds The Key

[citation needed]



"Only the fire department has access to that box"

[citation needed]



"And only we have a key to open it"

1-855-GET-KNOX

knoxhomebox.com

05/29/2016 04:08:13 AM

CH9



05/29/2016 04:08:13 AM

CH9



Sections

News

Weather

Good Day

FOX 7 Features

Contests

Traffic

Local

Texas US & World Politics Business Health Entertainment Strange Trending

'Knox Box' master key theft compromises thousands of buildings across Austin

Phoenix Fire Department Loses 850 Keys, Which Access Thousands of Local Businesses and Apartments



Phoenix Fire Department Loses 850 Keys, Which Access Thousands of Local Businesses and Apartments



The locks



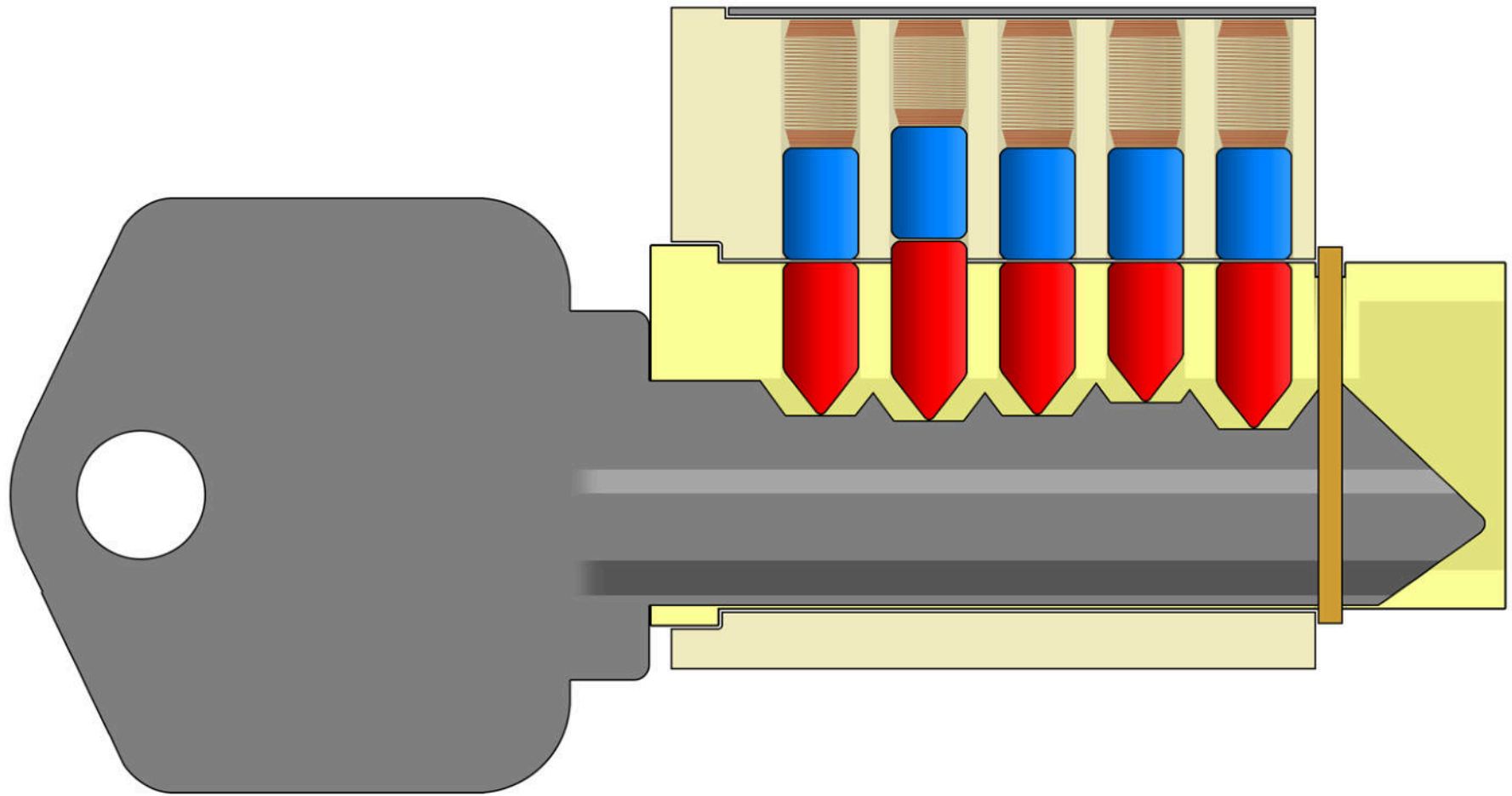
The locks

medeco[®]
HIGH SECURITY LOCKS

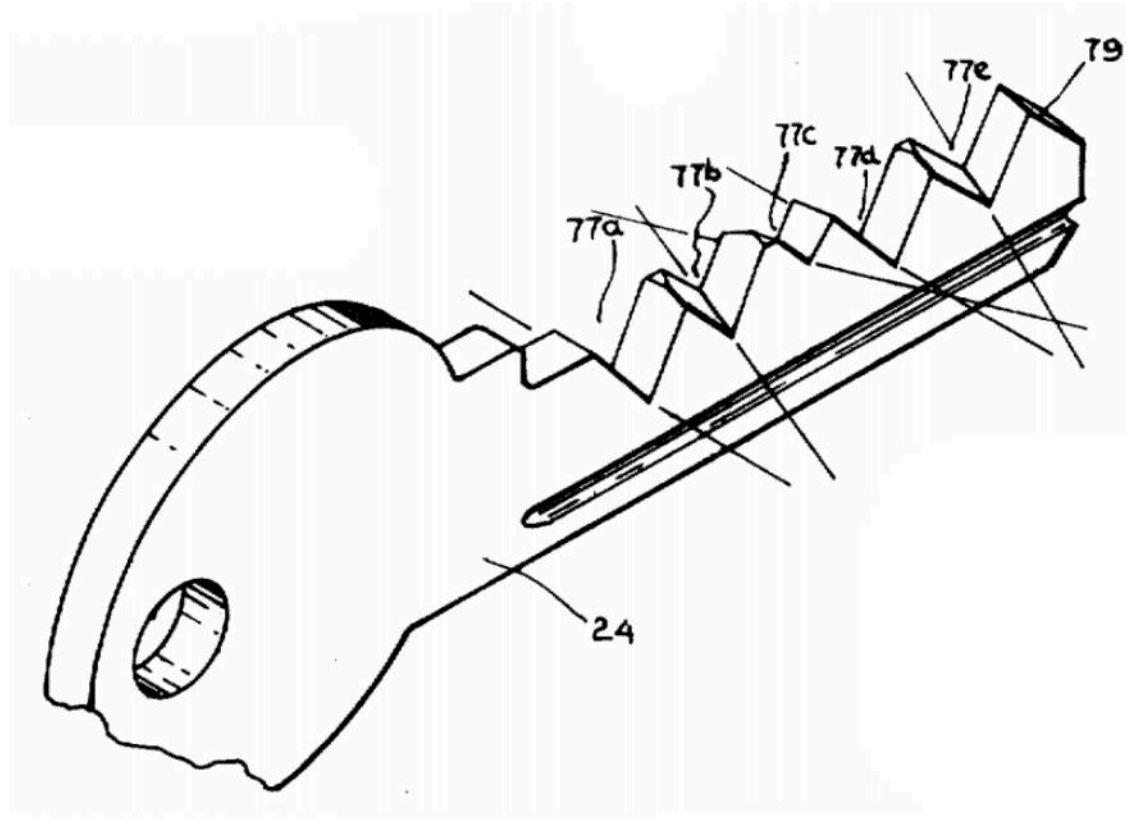
ASSA ABLOY



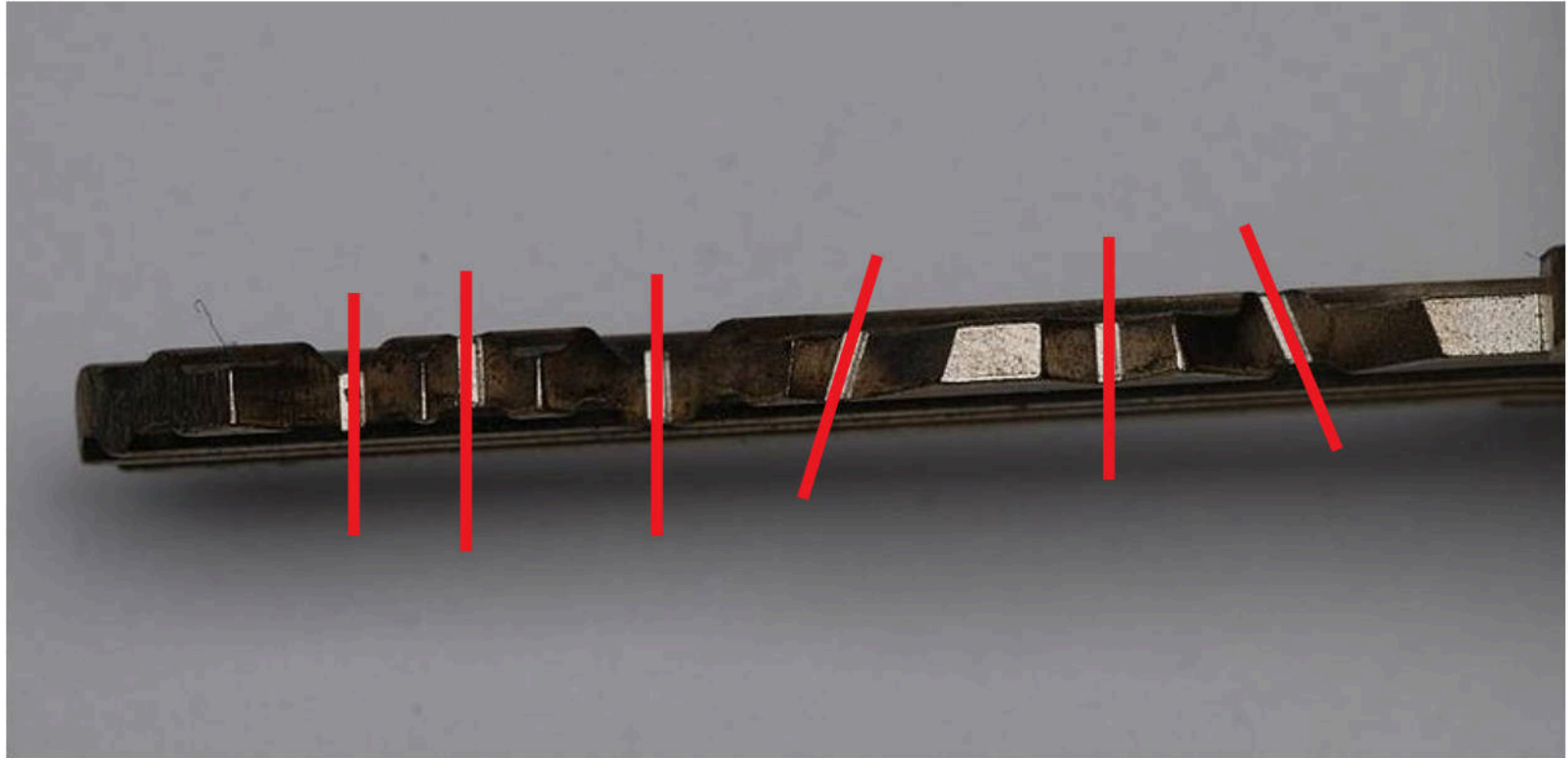
The locks



The locks



The locks



The locks



The locks



UL 1037



UL 437

The locks



UL 1037



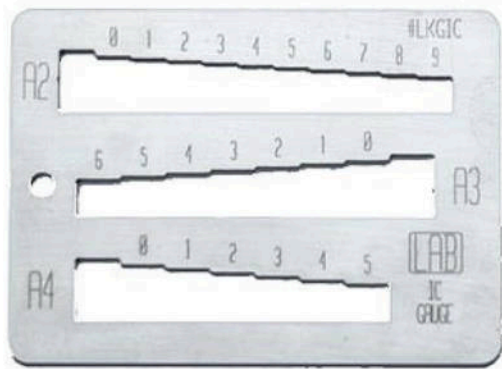
UL 437

Key control



Knox KeySecure

Key control



Key control

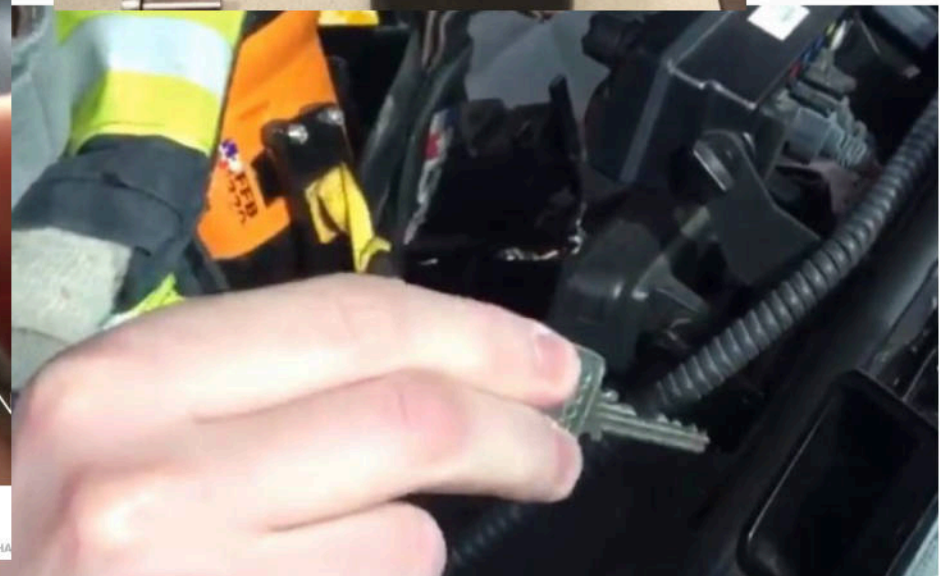
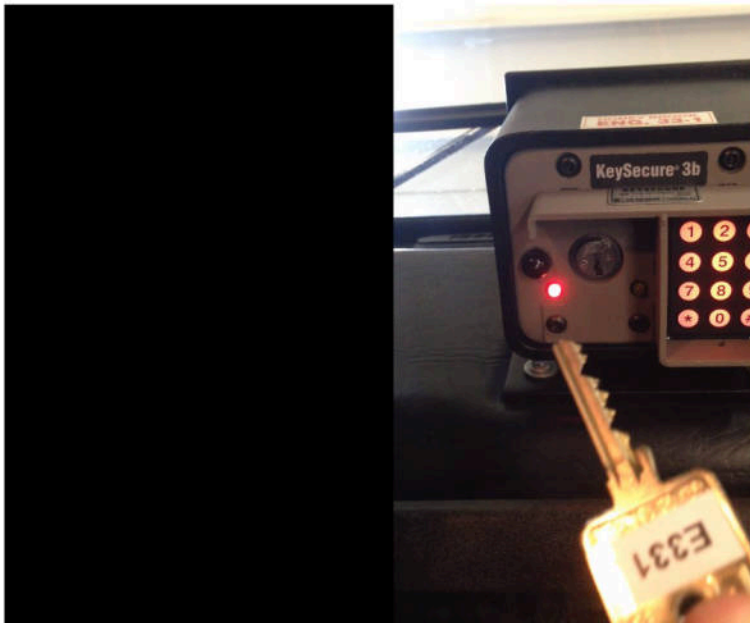


Key control



"Key control"

YouTube Search



Belltown FD—Knox Box Release Training Video

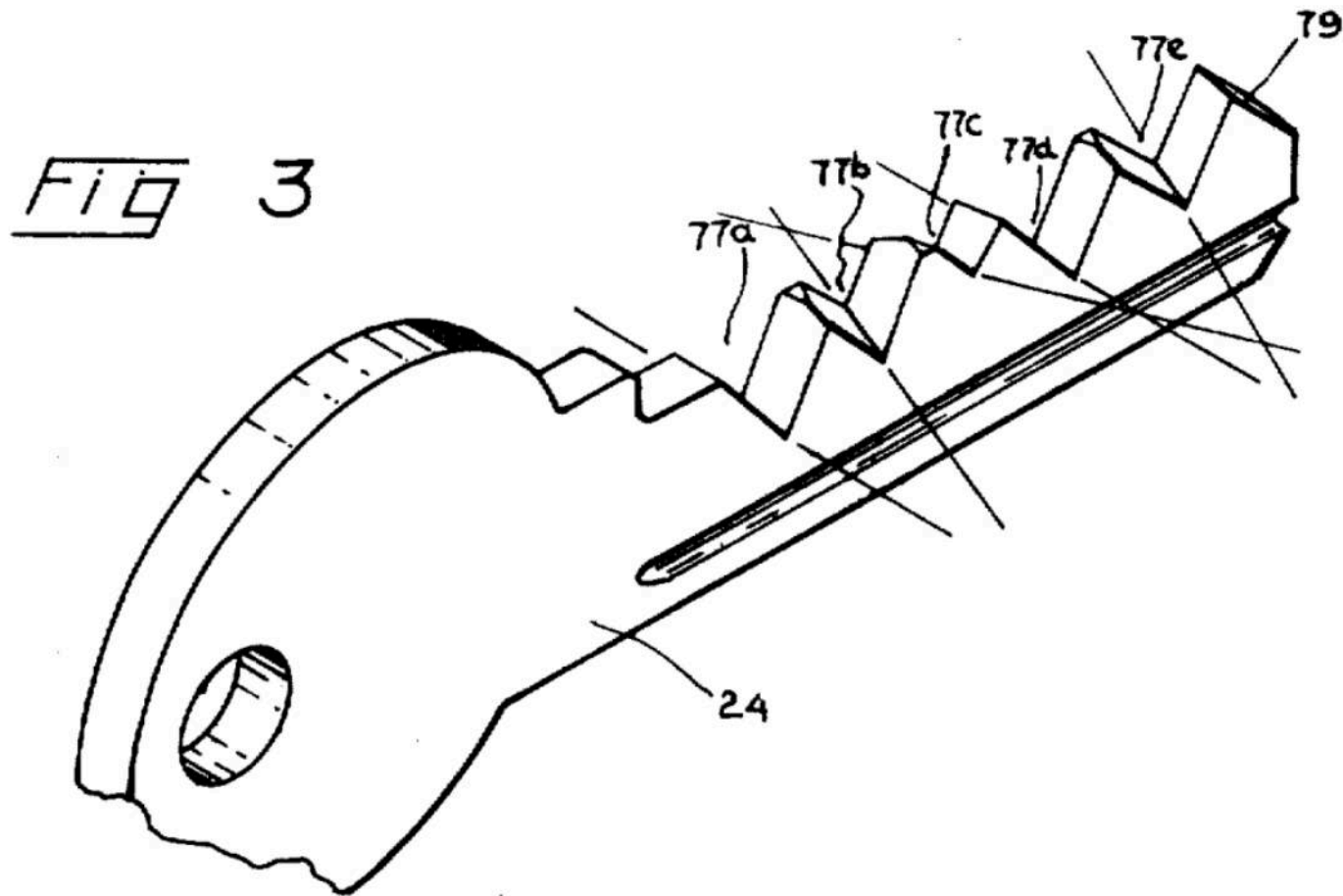
784 views

3 0 SHA

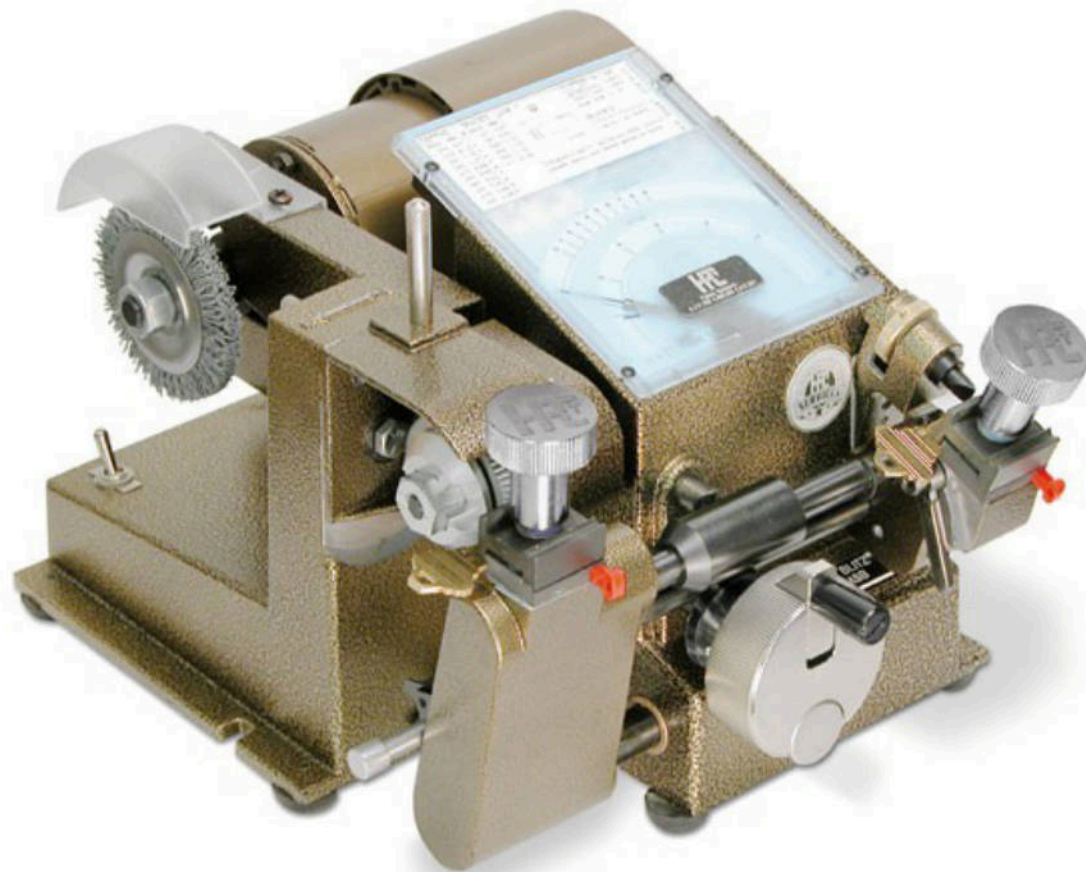
Key control



Key control?



Cutting keys

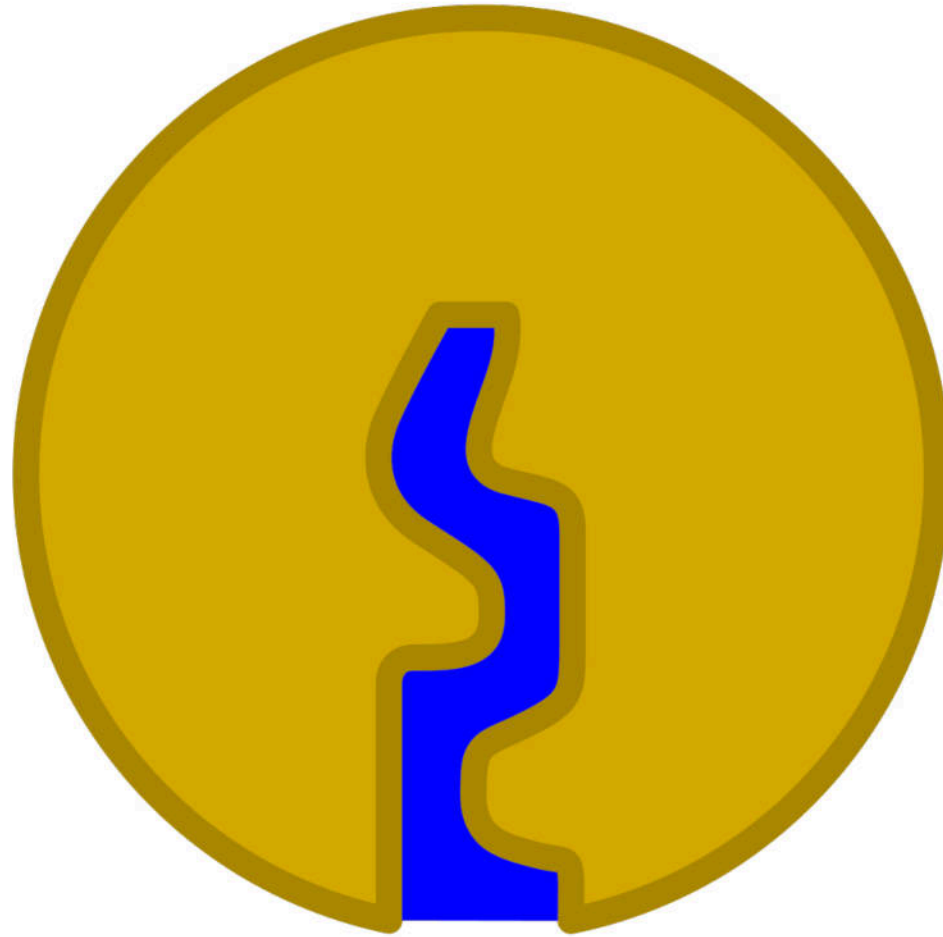


HPC Blitz

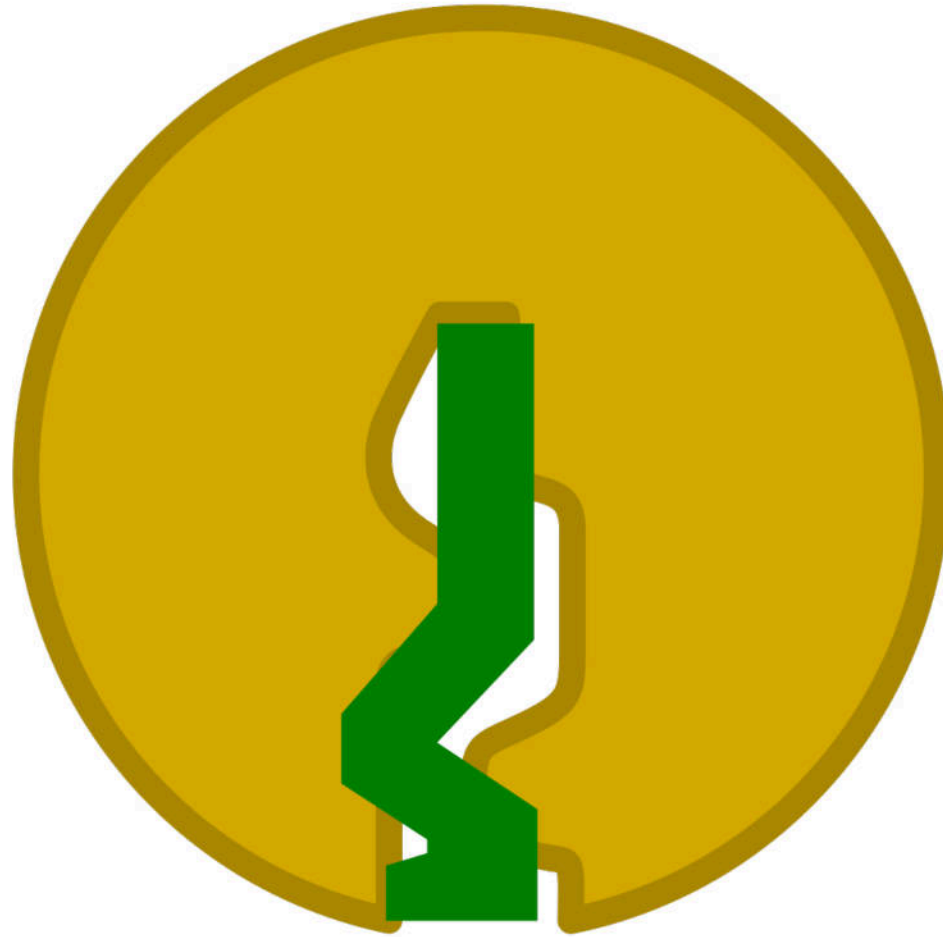
Blanks



Making blanks



Making blanks



Making blanks



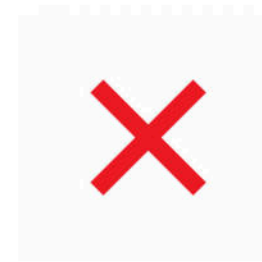
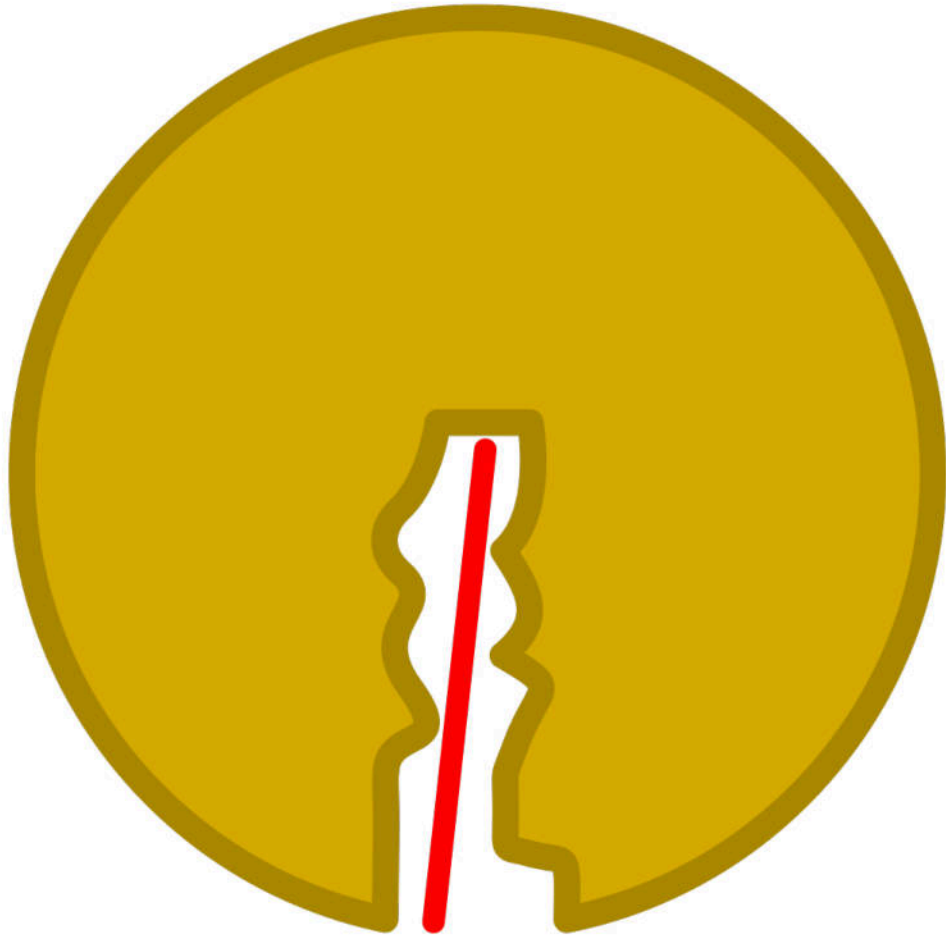
Making blanks



Making blanks



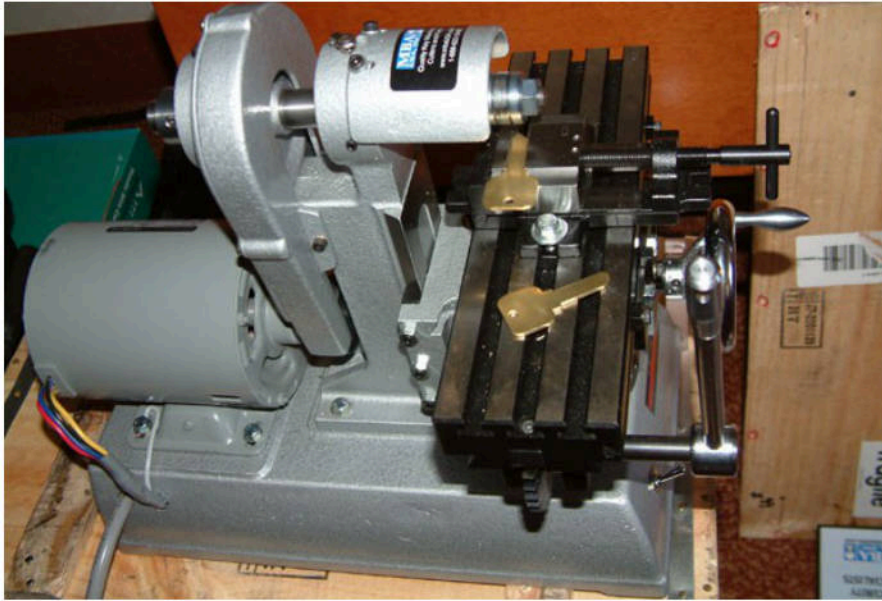
Making blanks



Making blanks



Making blanks



Keyway King



Easy Entrie

3d printing

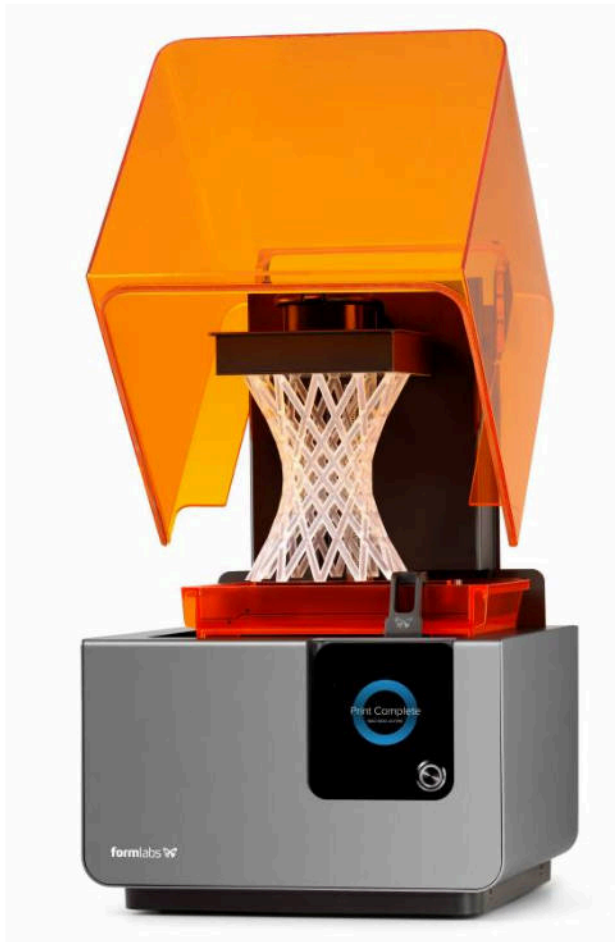


3d printing



3d printing

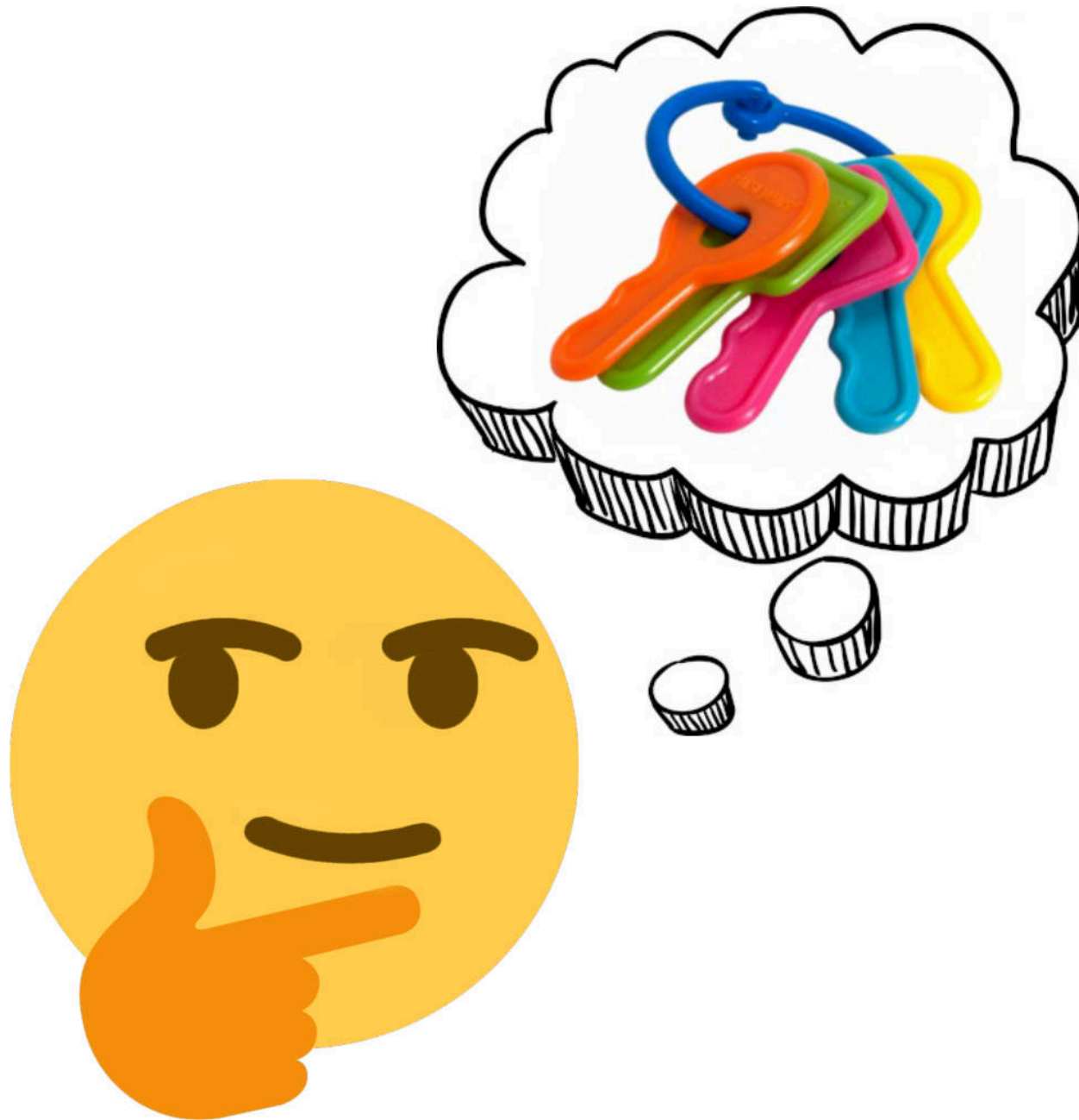
shapeways[★]



Hmmm



Hmmm



Key control



US006945082B2

(10) **Patent No.:** **US 6,945,082 B2**
(45) **Date of Patent:** **Sep. 20, 2005**

Key control



Key control



Key control?



Investigating the box



Investigating the box



Investigating the box



Investigating the box



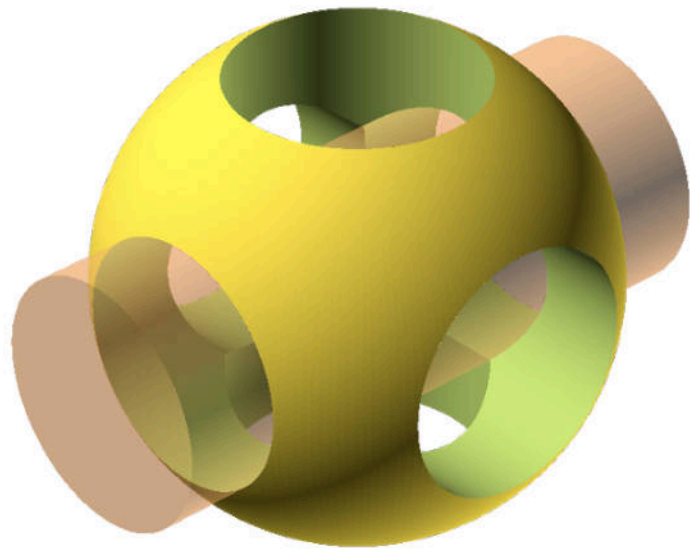
Investigating the box



Investigating the box



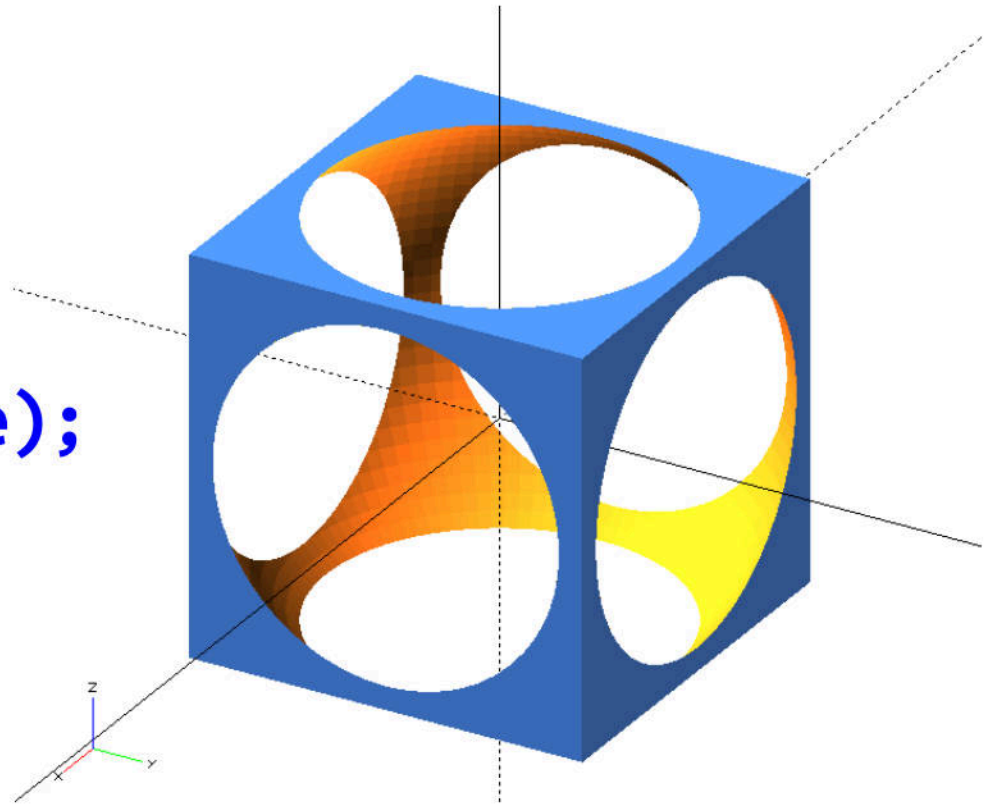
OpenSCAD



OpenScad

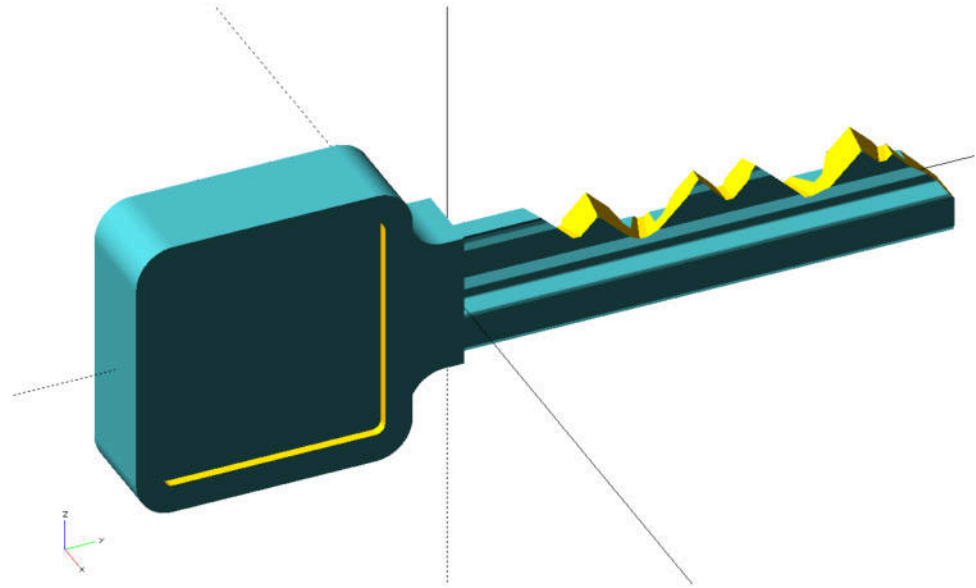
OpenSCAD

```
$fn = 100;  
difference() {  
    cube(3, center=true);  
    sphere(2);  
}
```

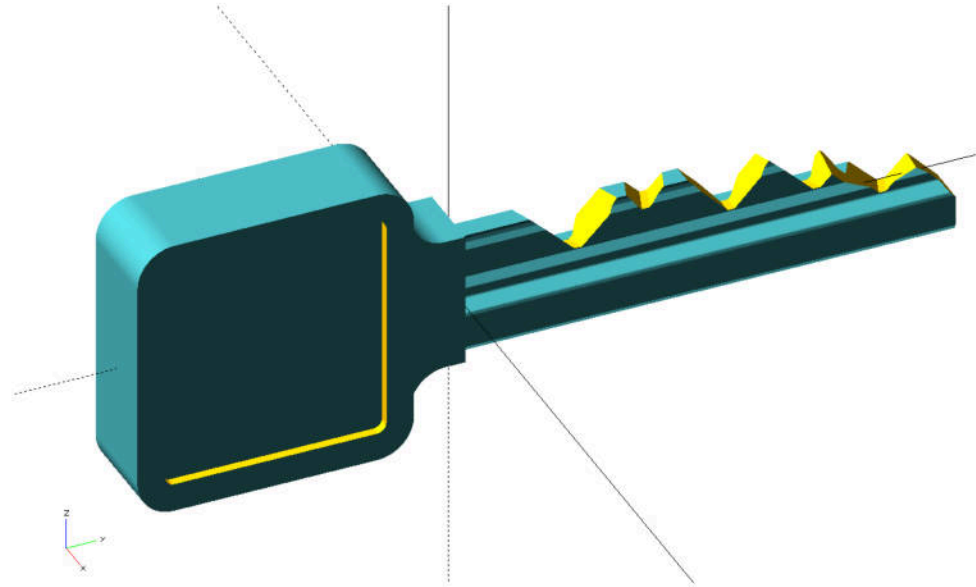


OpenSCAD

```
depths(2,4,2,4,2);
```



```
depths(3,1,3,3,7);
```



Prior art

- Nirav Patel
 - eclecti.cc
- Christian Holler
 - [autokey3d](http://autokey3d.com)
- Eric Van Albert
 - keygen.co

Modeling the keyway



Modeling the keyway



Modeling the keyway



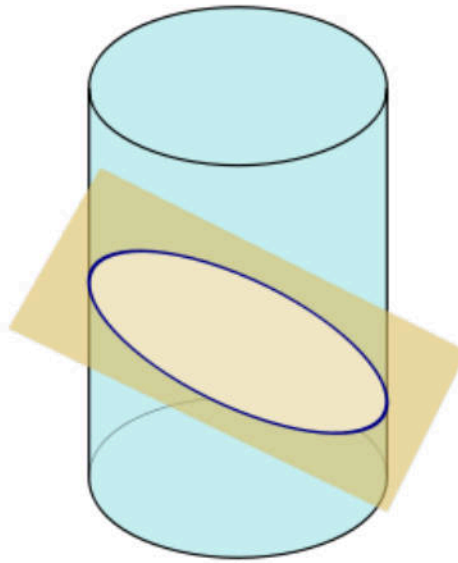
Modeling the keyway



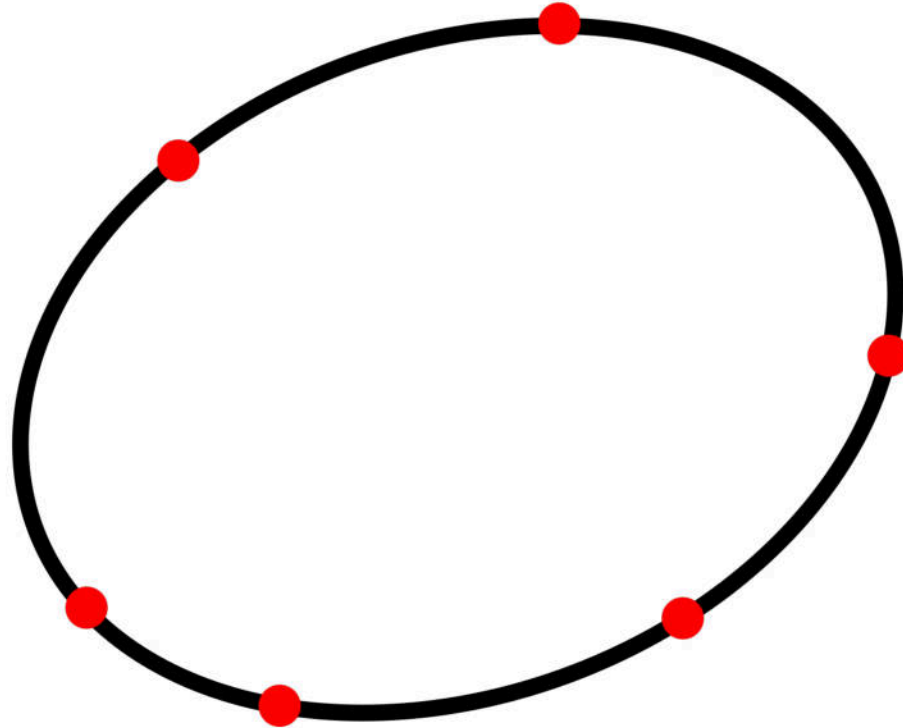
Modeling the keyway



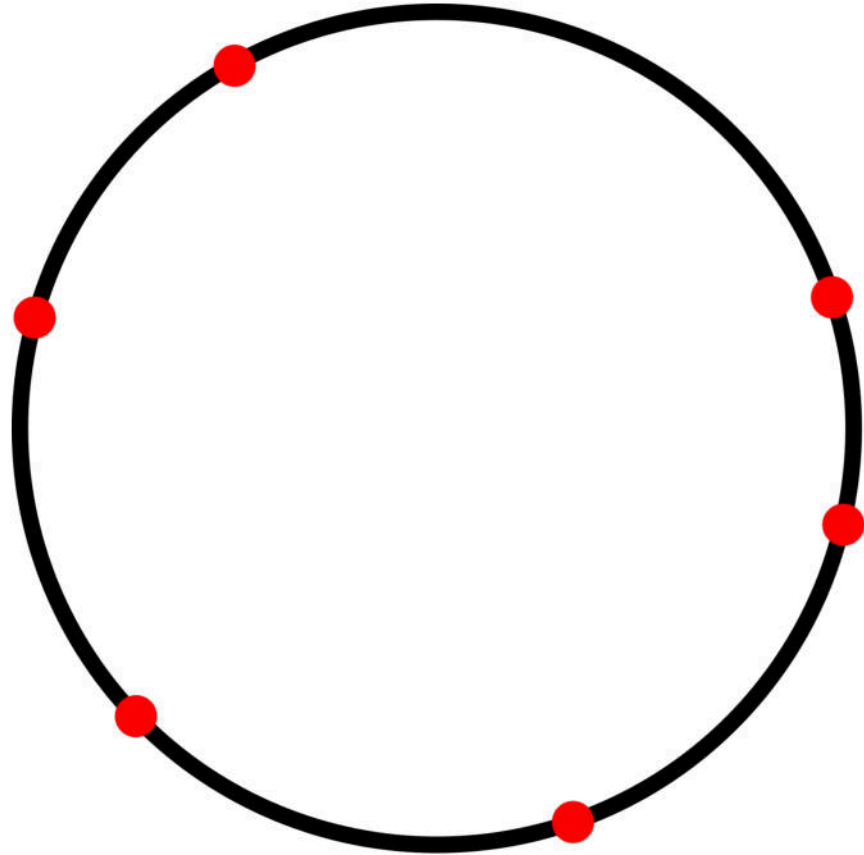
Modeling the keyway



Modeling the keyway



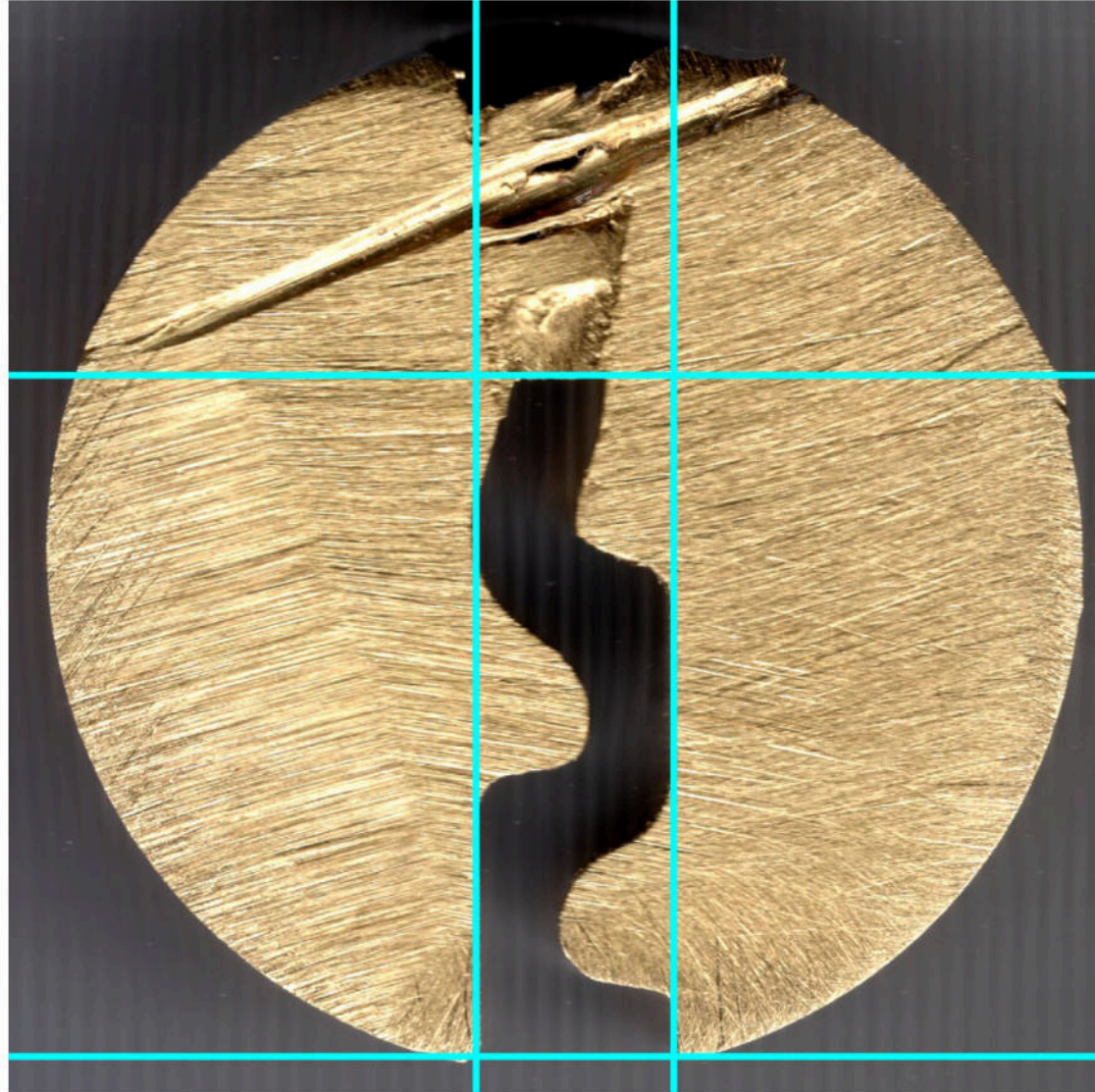
Modeling the keyway



Modeling the keyway



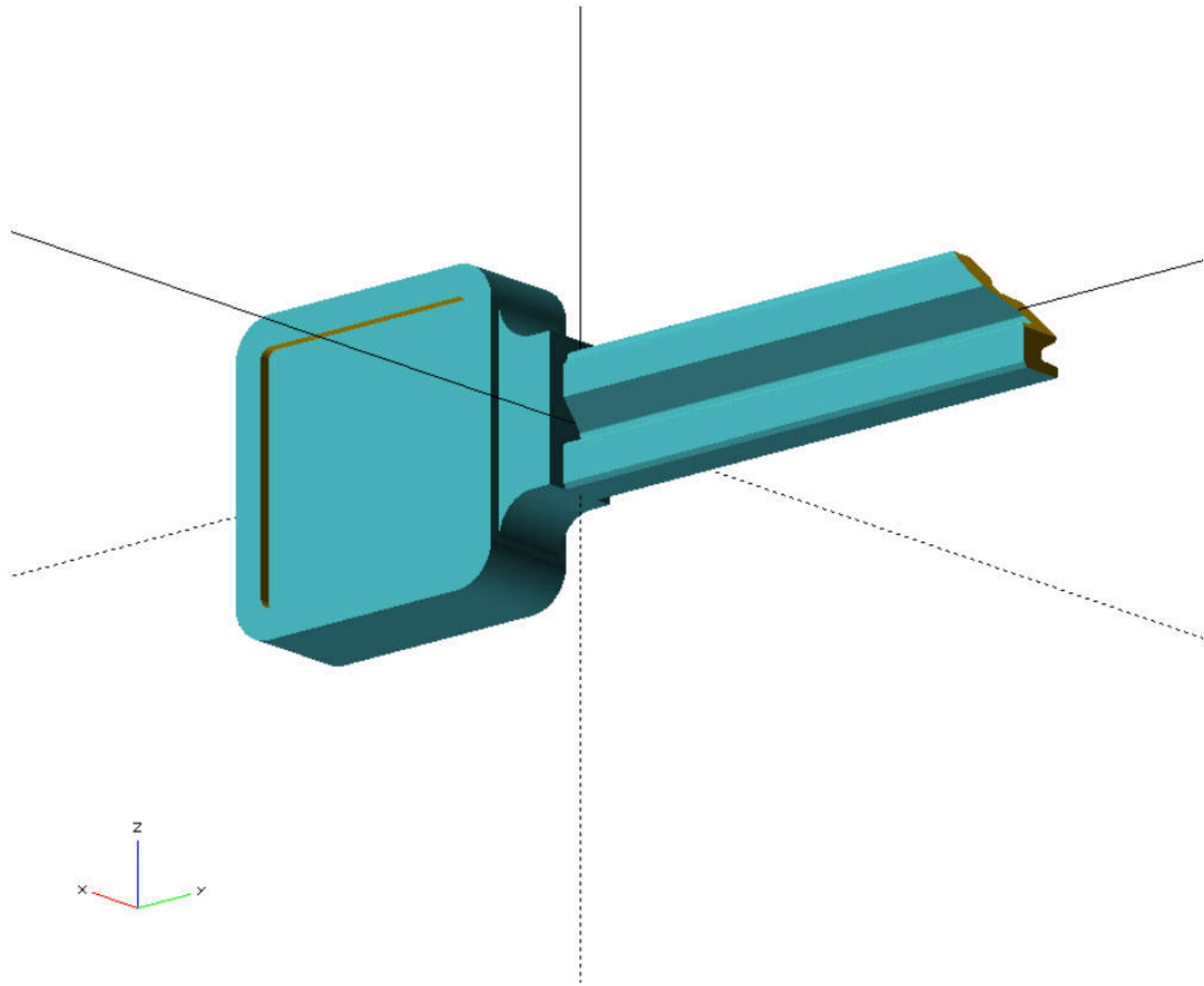
Modeling the keyway



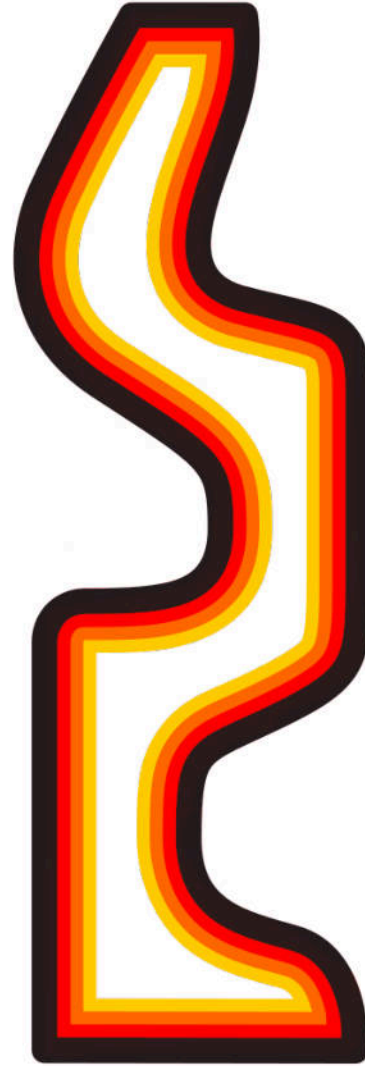
Modeling the keyway



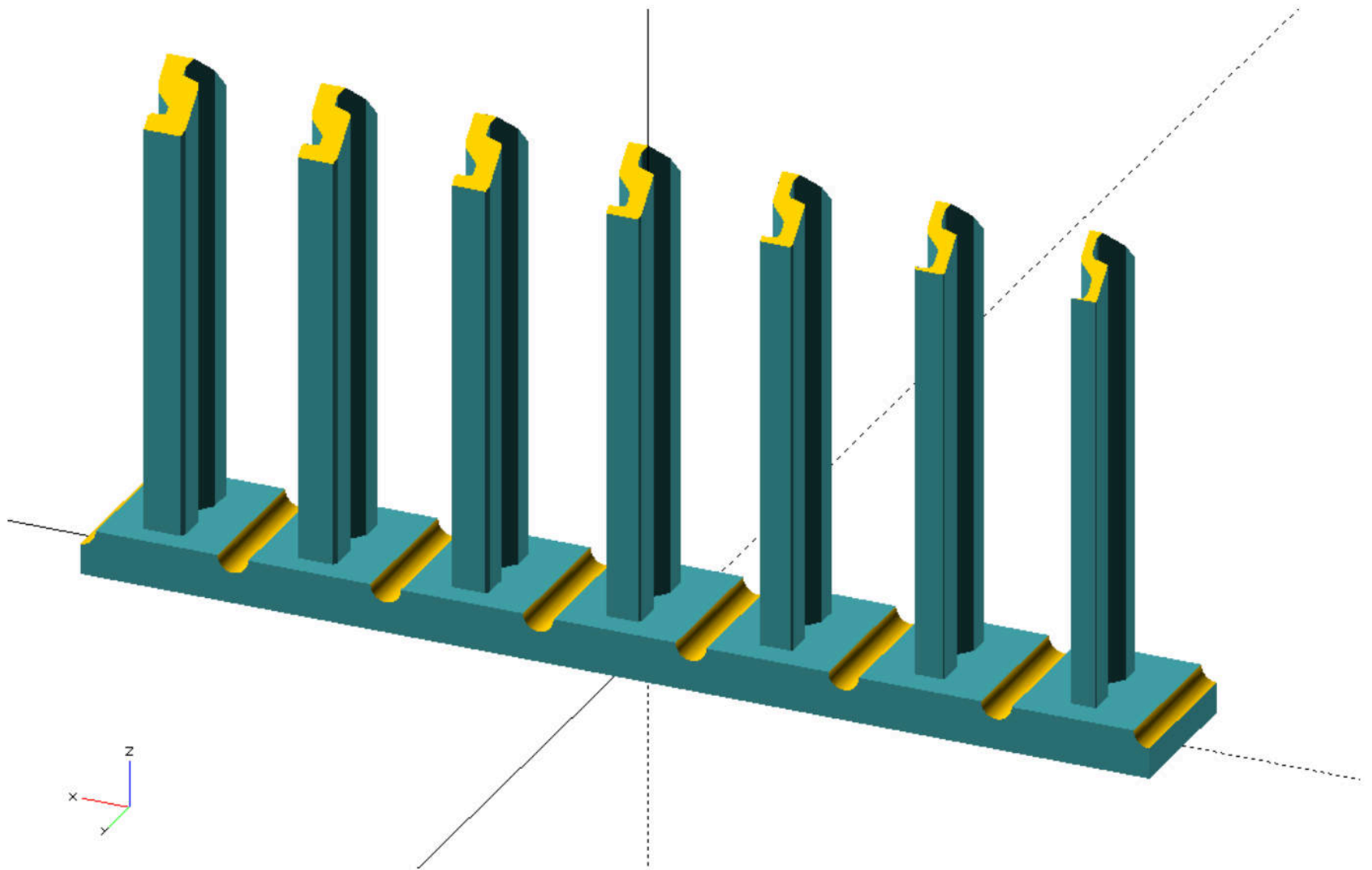
Modeling the keyway



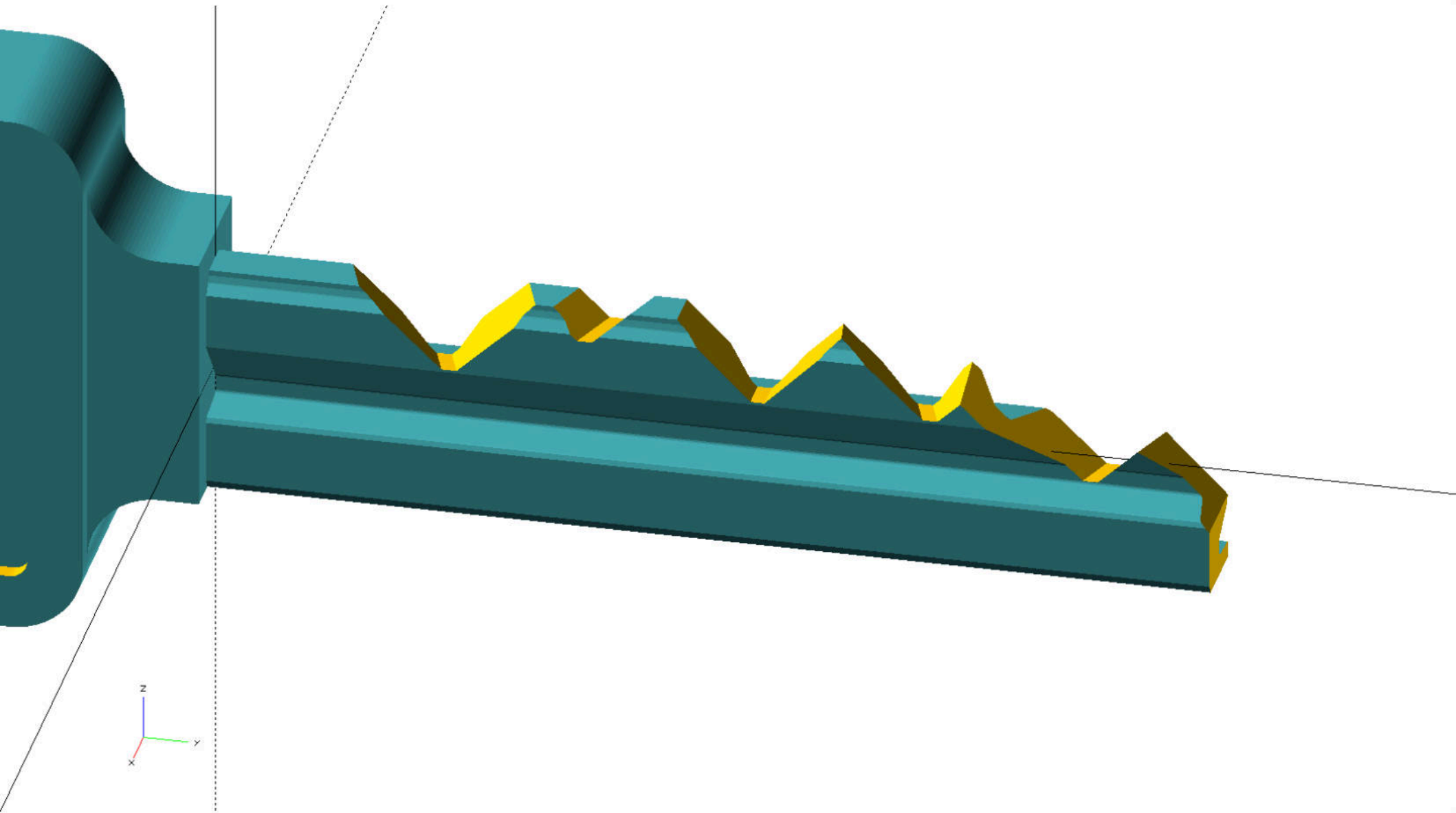
Modeling the keyway



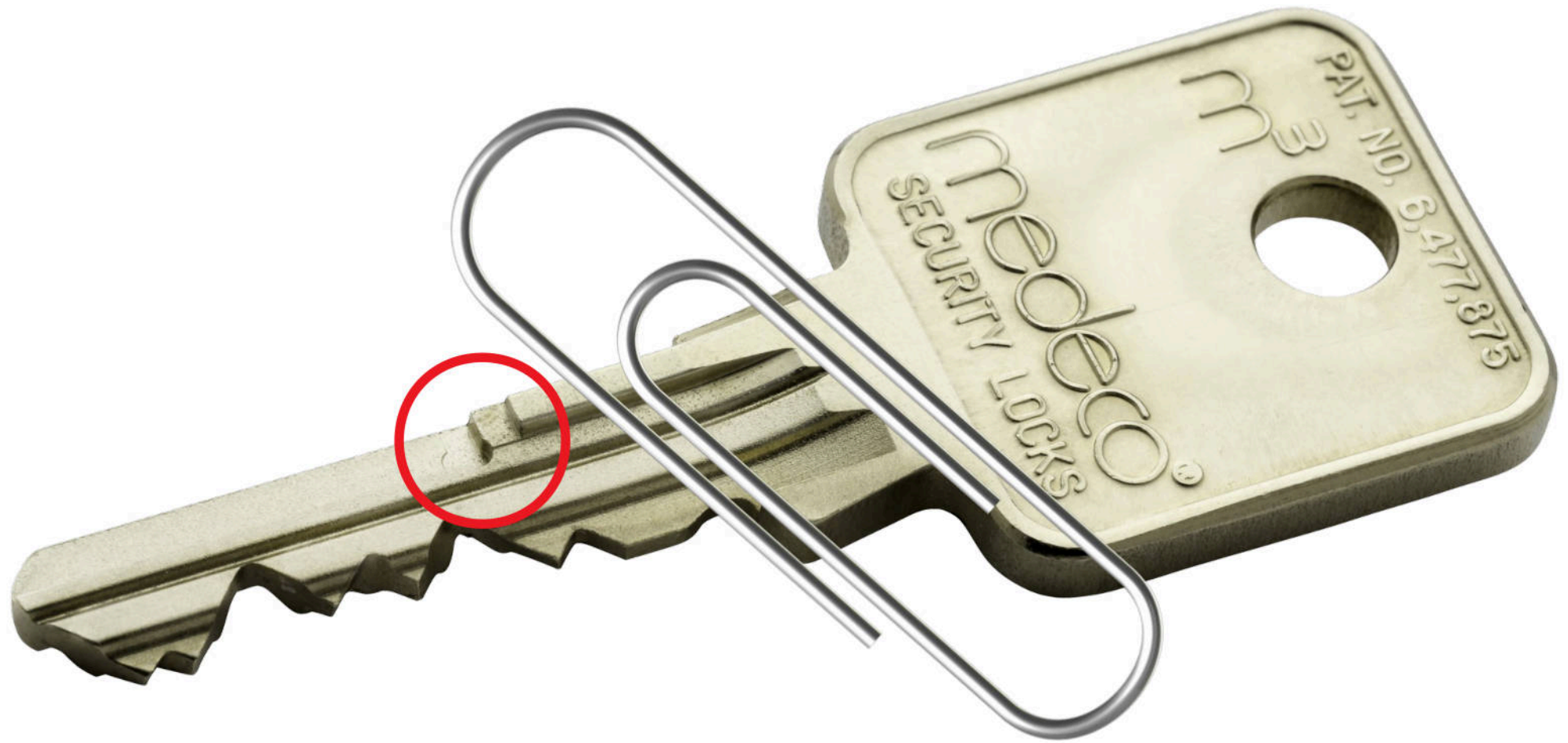
Modeling the keyway

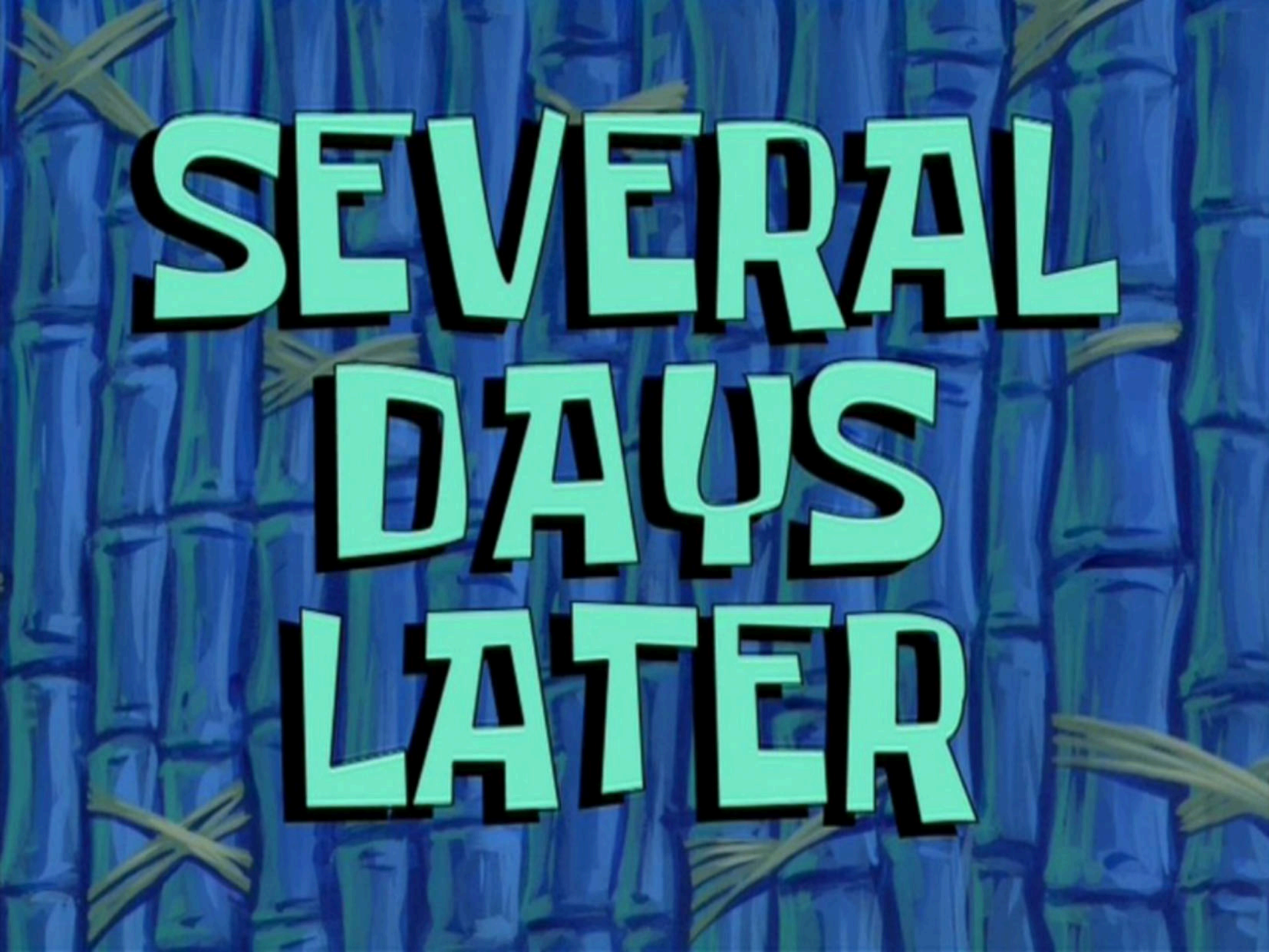


Modeling the cuts



M3 element



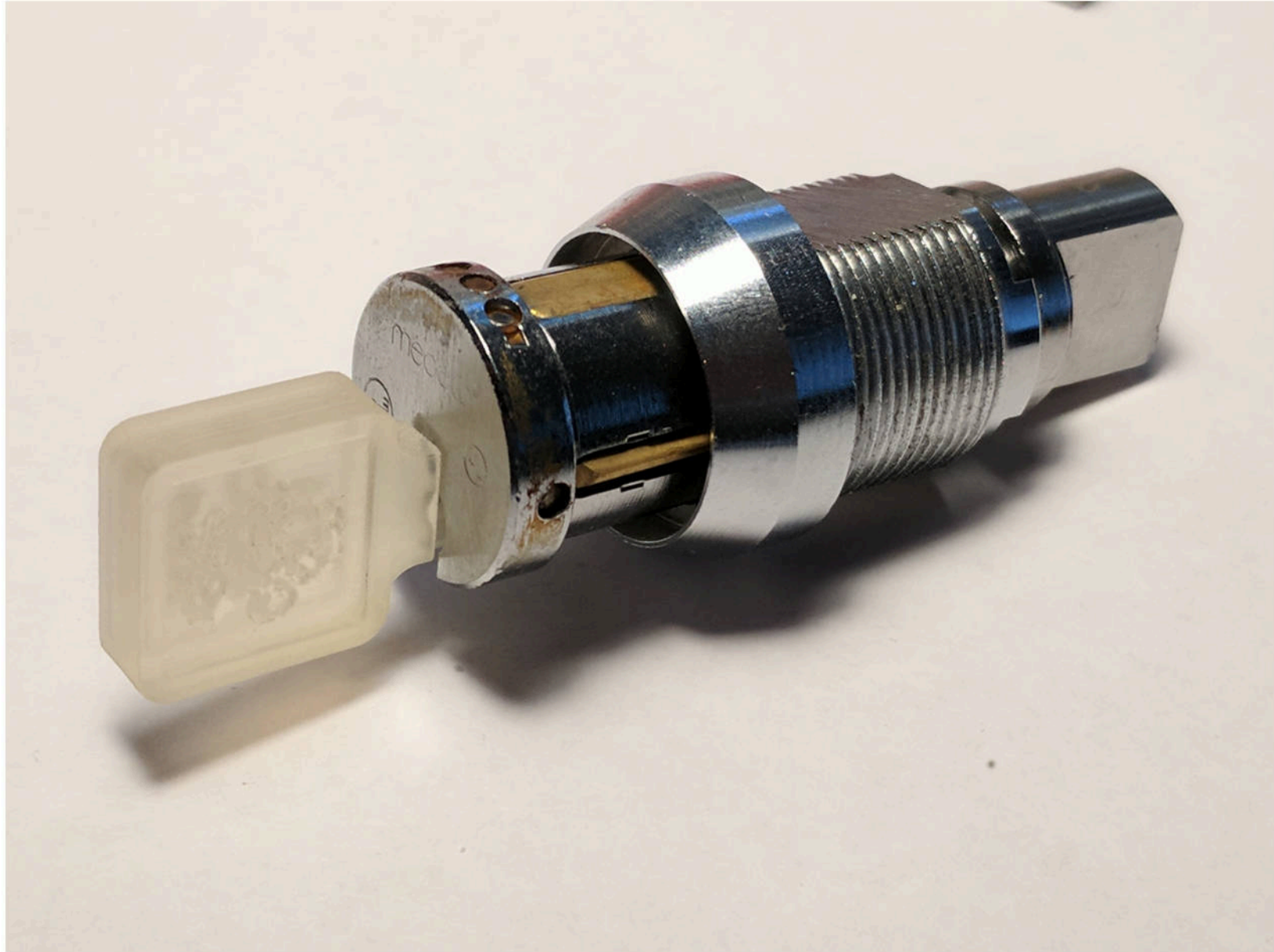


**SEVERAL
DAYS
LATER**

Pwned



Pwned



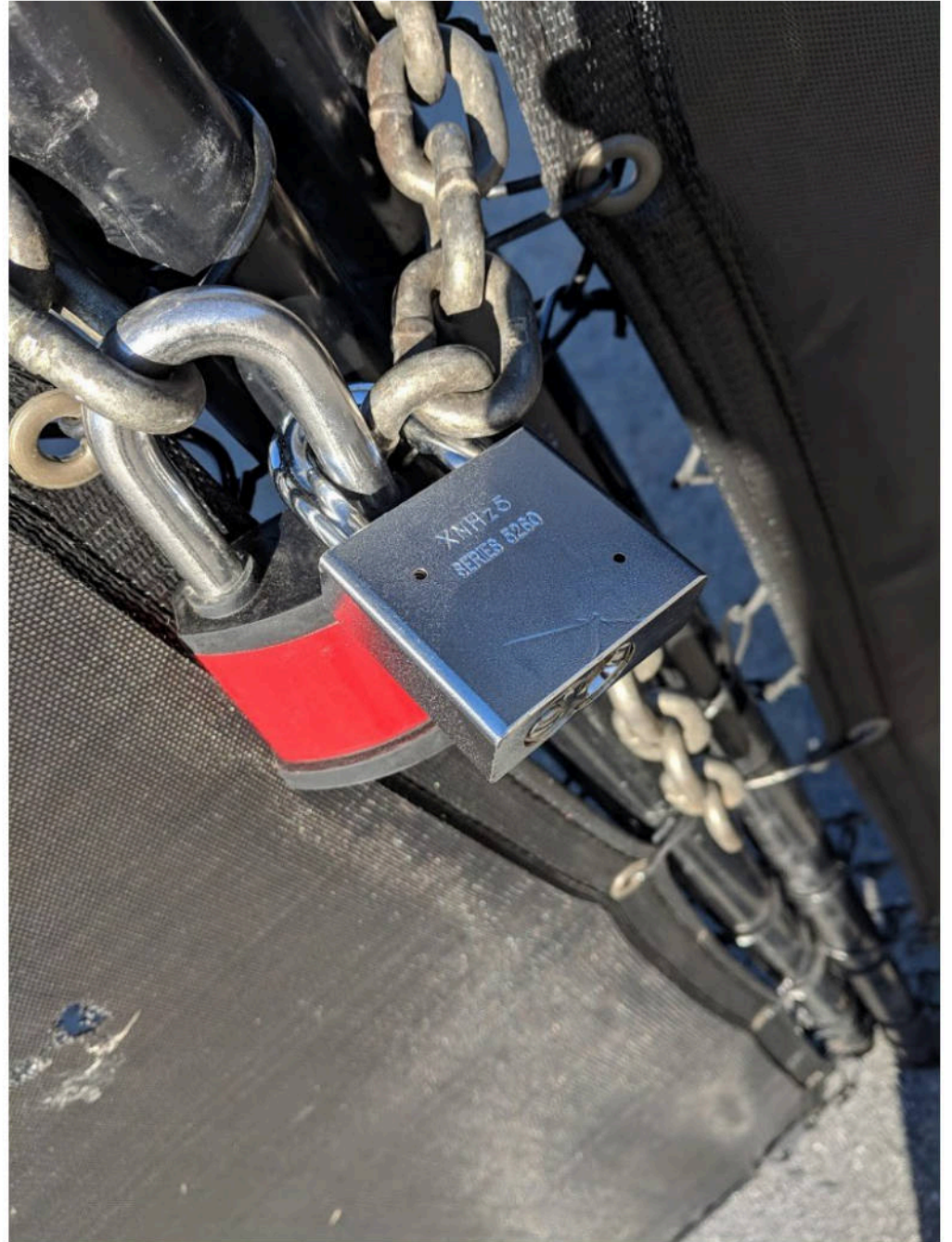
Sturdy



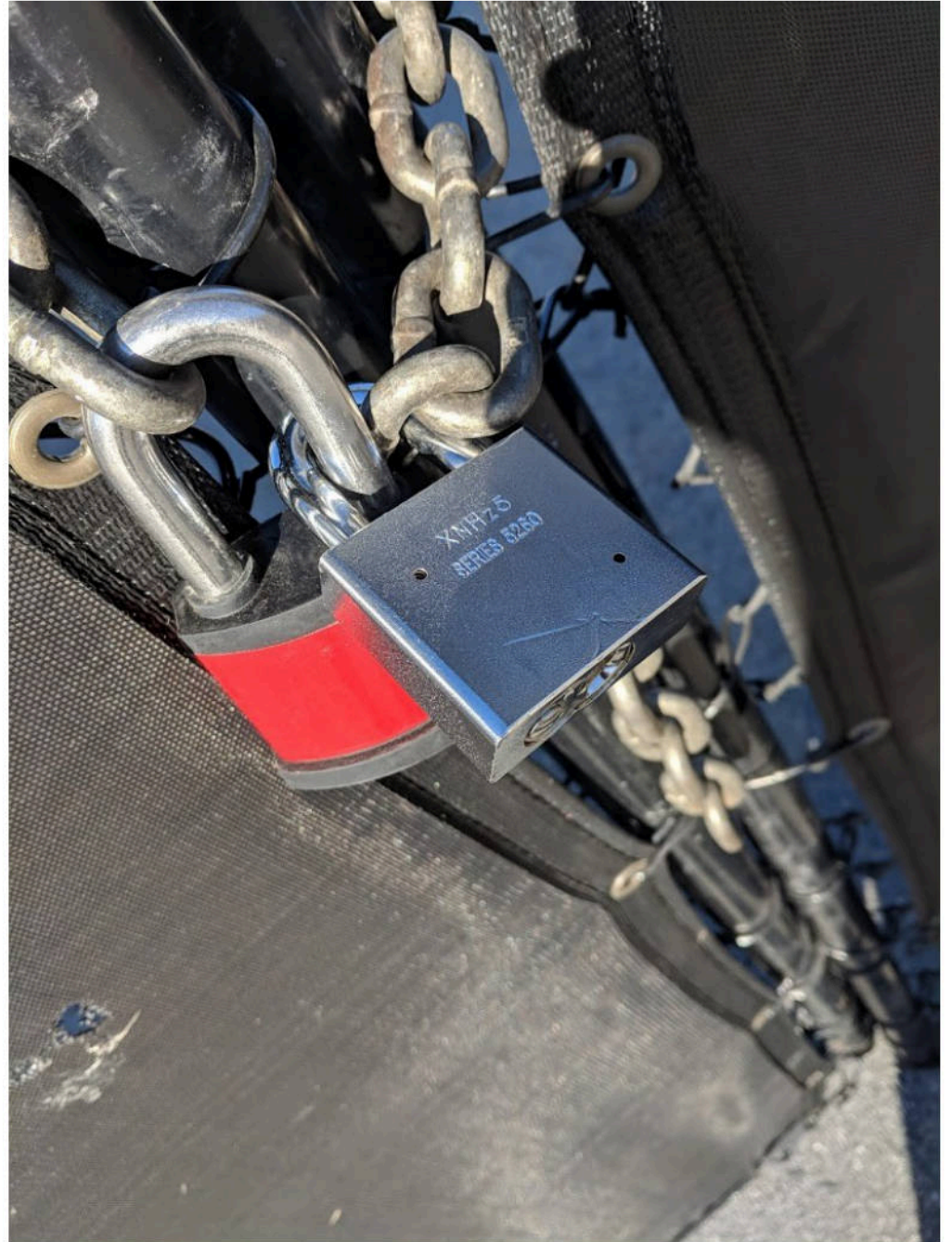
Very difficult to steal



Very difficult to steal



Very difficult to steal

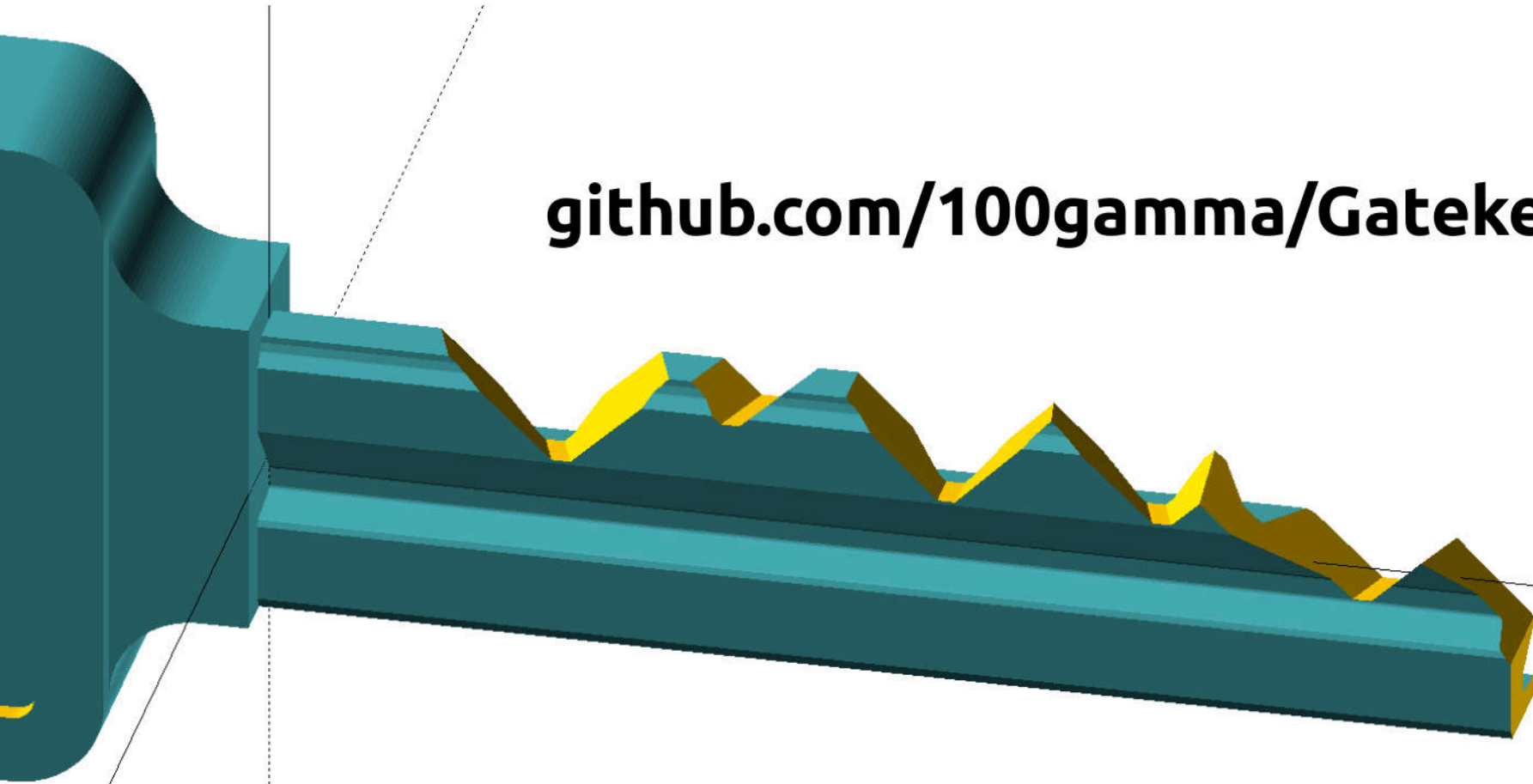


Very difficult to steal



Gatekeyper

github.com/100gamma/Gatekeyper



Change the locks?



Electromechanical locks



Electromechanical locks



Electromechanical locks



Combination locks

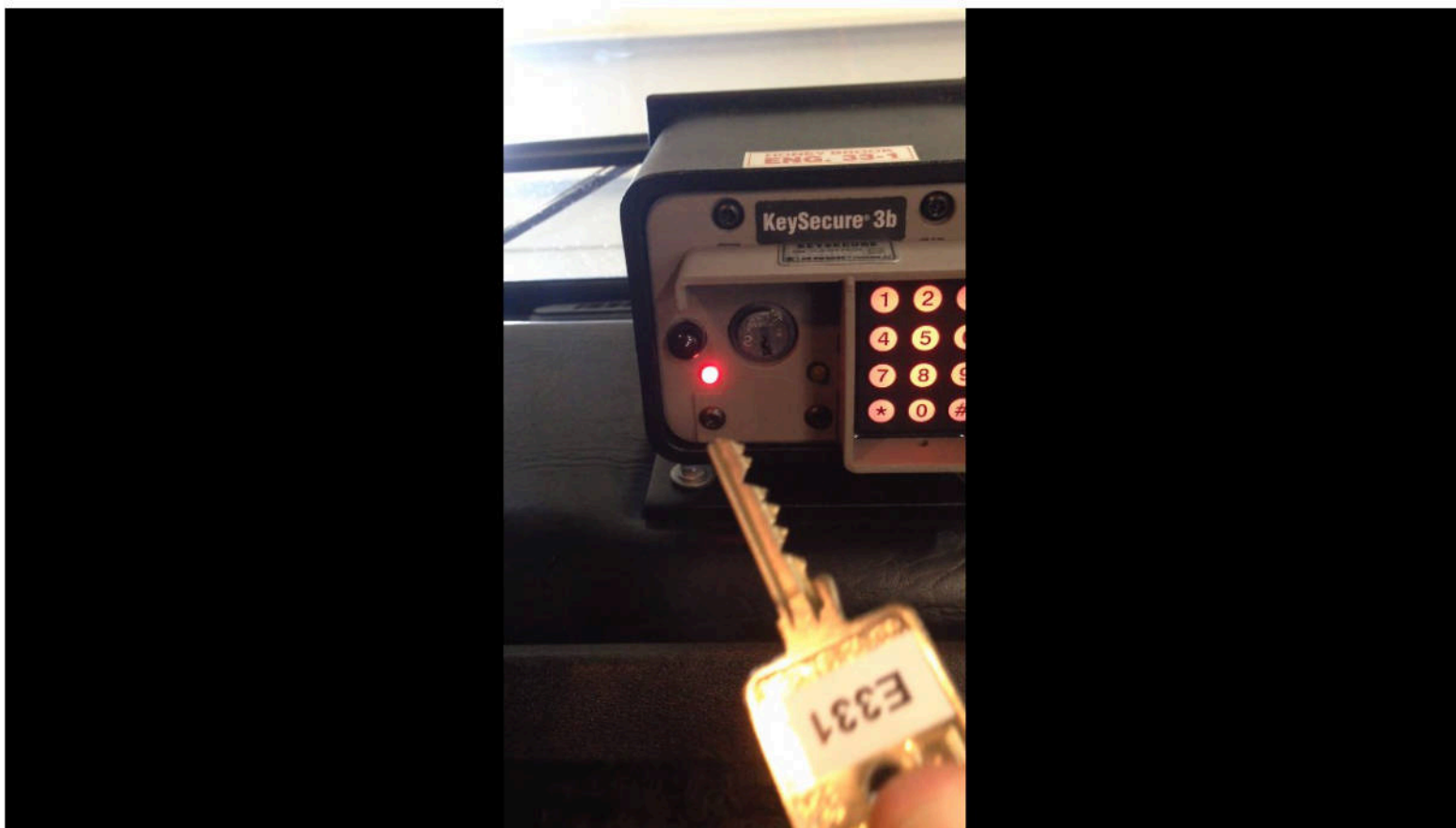


Stanton Concepts Robo Key System



YouTube

Search



Knox Box Tutorial

4,386 views

11

1

SHARE



Robert Klinger Jr.

Published on Mar 18, 2015

SUBSCRIBE 350

Category

Nonprofits & Activism

License

Standard YouTube License

Knox eLock



[CITATION NEEDED]



"The eLock cannot be hacked"

[CITATION NEEDED]



<https://www.youtube.com/watch?v=TC34dDzSdus>

Highly Secure

- AES encrypted
- eLock cores cannot be hacked or detected via RF or Bluetooth technology

Short-term solutions

- Tamper alert

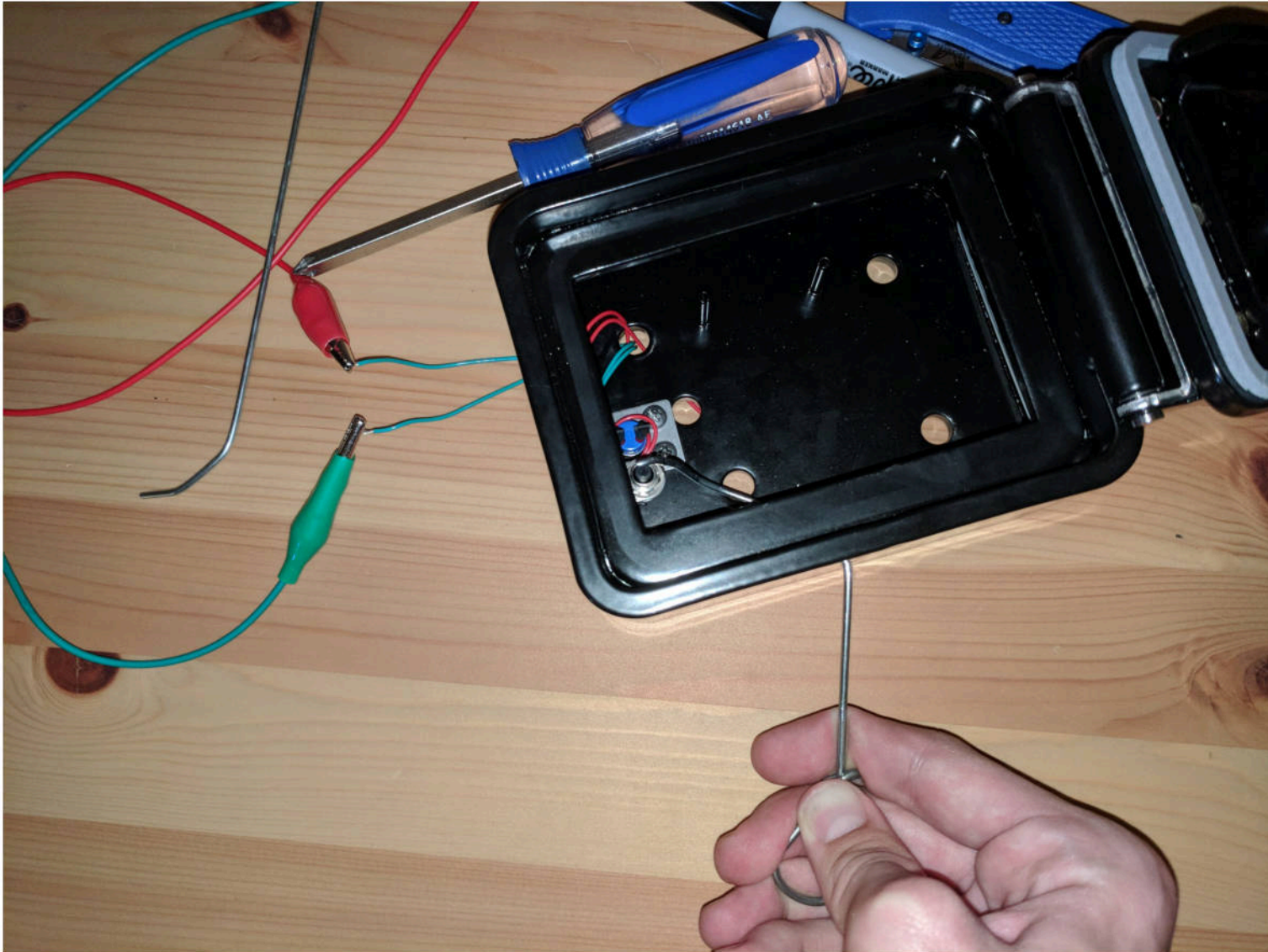
Tamper alert



Tamper alert?



Tampering with the tamper alert



Tampering with the tamper alert



Tampering with the tamper alert



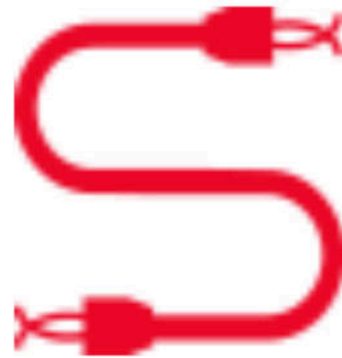
Tampering with the tamper alert



Tampering with the tamper alert



[citation needed]



Highest security when equipped
with optional alarm tamper alert

Short-term solutions

- Tamper alert
 - No drain hole
- Cameras
- Alarm

Long-term solutions

- Switch to a better product lol

Appendix

- github.com/100gamma/Gatekeyper

Thanks





\end{Talk}

 @m010ch_

zamak@protonmail.ch

github.com/100gamma/Gatekeyper